Supplementary Information

Two-Stage Synthetic-to-Real Transfer Learning for Automated Mammography Report Generation Using Vision-Language Models

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Supplementary Figure S1

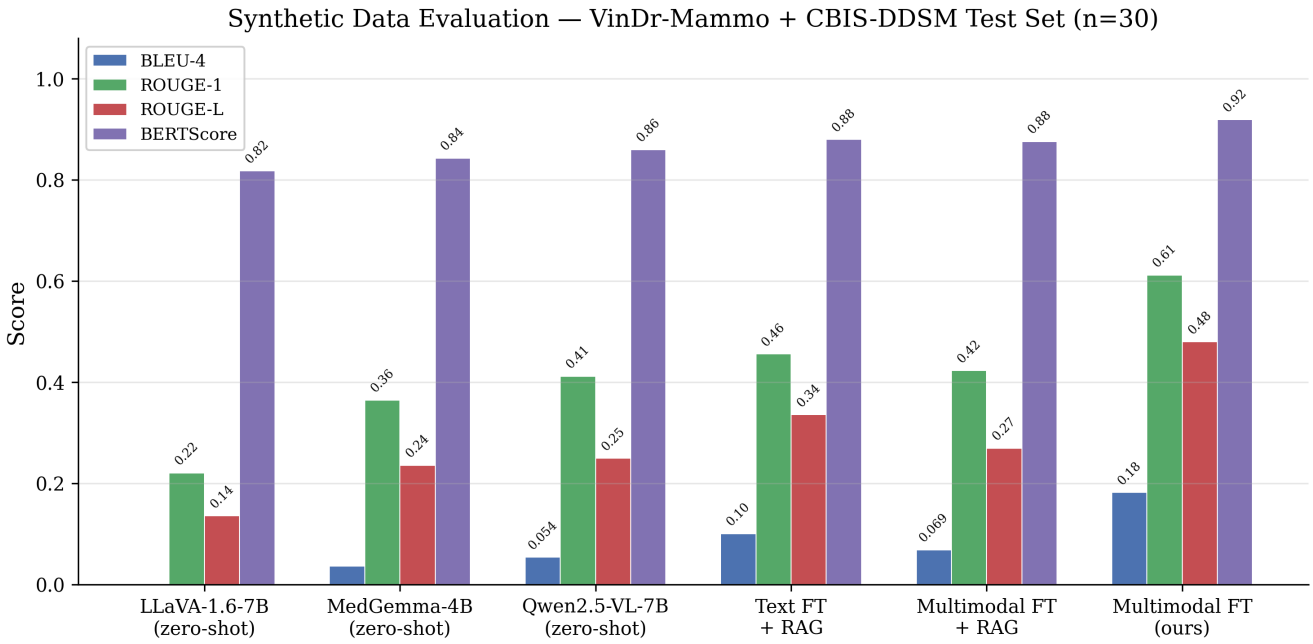


Figure S1: Stage 1 evaluation on synthetic test data. Multimodal fine-tuning without RAG achieves the best NLP metrics across all measures.

Supplementary Table S1

Each entry gives the image number and the split it belongs to. Entries marked † are the three test images whose reference report also occurs verbatim in the training split; they are excluded from the primary evaluation throughout the paper, which is reported on the remaining 49 test images, and the full 52-image split is retained as a sensitivity analysis. Totals: 407 train, 51 validation, 52 test (49 after exclusion).

Table S1: Split assignment of all 510 DMID images. Entries marked † are the three test images whose reference report also occurs verbatim in the training split.

001 train	002 train	003 train	004 train	005 train	006 train
007 train	008 train	009 train	010 train	011 train	012 train
013 train	014 train	015 train	016 train	017 train	018 train
019 train	020 train	021 train	022 train	023 train	024 train
025 train	026 train	027 train	028 train	029 train	030 train
031 train	032 train	033 train	034 train	035 train	036 train
037 train	038 train	039 train	040 train	041 train	042 train
043 train	044 train	045 train	046 train	047 train	048 train
049 train	050 train	051 train	052 train	053 train	054 train
055 train	056 train	057 train	058 train	059 train	060 train
061 train	062 train	063 train	064 train	065 train	066 train
067 train	068 train	069 train	070 train	071 train	072 train
073 train	074 train	075 train	076 train	077 train	078 train
079 train	080 train	081 train	082 train	083 train	084 train
085 train	086 train	087 train	088 train	089 train	090 train
091 train	092 train	093 train	094 train	095 train	096 train
097 train	098 train	099 train	100 train	101 train	102 train
103 train	104 train	105 train	106 train	107 train	108 train
109 train	110 train	111 train	112 train	113 train	114 train
115 train	116 train	117 train	118 train	119 train	120 train
121 train	122 train	123 train	124 train	125 train	126 train
127 train	128 train	129 train	130 train	131 train	132 train
133 train	134 train	135 train	136 train	137 train	138 train
139 train	140 train	141 train	142 train	143 train	144 train
145 train	146 train	147 train	148 train	149 train	150 train
151 train	152 train	153 train	154 train	155 train	156 train
157 train	158 train	159 train	160 train	161 train	162 train
163 train	164 train	165 train	166 train	167 train	168 train
169 train	170 train	171 train	172 train	173 train	174 train
175 train	176 train	177 train	178 train	179 train	180 train
181 train	182 train	183 train	184 train	185 train	186 train
187 train	188 train	189 train	190 train	191 train	192 train
193 train	194 train	195 train	196 train	197 train	198 train
199 train	200 train	201 train	202 train	203 train	204 train
205 train	206 train	207 train	208 train	209 train	210 train
211 train	212 train	213 train	214 train	215 train	216 train
217 train	218 train	219 train	220 train	221 train	222 train
223 train	224 train	225 train	226 train	227 train	228 train
229 train	230 train	231 train	232 train	233 train	234 train
235 train	236 train	237 train	238 train	239 train	240 train
241 train	242 train	243 train	244 train	245 train	246 train
247 train	248 train	249 train	250 train	251 train	252 train
253 train	254 train	255 train	256 train	257 train	258 train
259 train	260 train	261 train	262 train	263 train	264 train
265 train	266 train	267 train	268 train	269 train	270 train
271 train	272 train	273 train	274 train	275 train	276 train
277 train	278 train	279 train	280 train	281 train	282 train

Supplementary Table S1 (continued)

283 train	284 train	285 train	286 train	287 train	288 train
289 train	290 train	291 train	292 train	293 train	294 train
295 train	296 train	297 train	298 train	299 train	300 train
301 train	302 train	303 train	304 train	305 train	306 train
307 train	308 train	309 train	310 train	311 train	312 train
313 train	314 train	315 train	316 train	317 train	318 train
319 train	320 train	321 train	322 train	323 train	324 train
325 train	326 train	327 train	328 train	329 train	330 train
331 train	332 train	333 train	334 train	335 train	336 train
337 train	338 train	339 train	340 train	341 train	342 train
343 train	344 train	345 train	346 train	347 train	348 train
349 train	350 train	351 train	352 train	353 train	354 train
355 train	356 train	357 train	358 train	359 train	360 train
361 train	362 train	363 train	364 train	365 train	366 train
367 train	368 train	369 train	370 train	371 train	372 train
373 train	374 train	375 train	376 train	377 train	378 train
379 train	380 train	381 train	382 train	383 train	384 train
385 train	386 train	387 train	388 train	389 train	390 train
391 train	392 train	393 train	394 train	395 train	396 train
397 train	398 train	399 train	400 train	401 train	402 train
403 train	404 train	405 train	406 train	407 train	408 val
409 val	410 val	411 val	412 val	413 val	414 val
415 val	416 val	417 val	418 val	419 val	420 val
421 val	422 val	423 val	424 val	425 val	426 val
427 val	428 val	429 val	430 val	431 val	432 val
433 val	434 val	435 val	436 val	437 val	438 val
439 val	440 val	441 val	442 val	443 val	444 val
445 val	446 val	447 val	448 val	449 val	450 val
451 val	452 val	453 val	454 val	455 val	456 val
457 val	458 val	459 test	460 test	461 test	462 test†
463 test	464 test†	465 test	466 test	467 test	468 test
469 test	470 test	471 test	472 test	473 test	474 test
475 test	476 test	477 test	478 test	479 test	480 test
481 test	482 test	483 test	484 test	485 test	486 test
487 test	488 test	489 test	490 test	491 test	492 test
493 test	494 test	495 test	496 test	497 test	498 test
499 test	500 test	501 test	502 test	503 test	504 test
505 test	506 test†	507 test	508 test	509 test	510 test

Supplementary Table S2

Table S2: Data efficiency: performance as a function of real training examples N. Δ denotes relative improvement of two-stage over DMID-only. **Protocol note.** This experiment was run under its own protocol—adapter rank $r = 16$, the training-matched prompt, and a single random initialisation—and per-sample generations were not retained, so confidence intervals cannot be computed for these rows, they cannot be recomputed on the leak-free test subset used elsewhere, and they are *not* comparable row-by-row with Table 4 of the main text. The table is internally consistent and is reported as a separate experiment. Reconciling it with Table 4 of the main text would require 18 further training runs (3 data volumes $\times$ 2 schemes $\times$ 3 initialisations), which we have not carried out.

	ROUGE-L		**ROUGE-1**		**BLEU-4**	
N	DMID-only	Two-stage	DMID-only	Two-stage	DMID-only	Two-stage
50	0.502	0.522 (+4.0%)	0.604	0.591	0.093	0.121
100	0.542	0.570 (+5.1%)	0.618	0.644	0.103	0.135
200	0.560	0.599 (+7.0%)	0.639	0.665	0.149	0.184
407	0.615	**0.662** (+7.6%)	0.686	**0.724**	0.221	**0.307**

Supplementary Table S3

Table S3: ACR breast density on the leak-free DMID test subset ($n = 49$), moved from Table 6 of the main text. Accuracy is exact agreement with the density category stated in the reference report; κ is Cohen's kappa. The reference distribution is A 3, B 22, C 19, D 5. Three of the four fine-tuned models assign a single category to every case—ACR C for DMID-only at $r = 16$, ACR B for both $r = 64$ models—so their accuracy equals the prevalence of that category and $\kappa = 0$: these values reflect a constant prediction (mode collapse), not density perception, and they cannot be read as evidence that the models assess density. The one configuration with non-zero agreement (two-stage, $r = 16$) is also the one with the lowest BI-RADS agreement in Table 6. The accuracy of the $r = 64$ models equals the prevalence of ACR B in the test subset (0.449) because they assign ACR B to every case; this figure must not be read as evidence of density perception ($\kappa = 0$).

Method	n	**Acc.**	κ	**Predicted categories**
MedGemma-4B (ZS)	17[†]	0.176	−0.026	A 15, B 1, C 1
DMID-only, r=16	49	0.388	0.000	C 49
Two-stage, r=16	49	0.490	0.108	B 37, C 11, D 1
DMID-only, r=64	49	0.449	0.000	B 49
Two-stage, r=64	49	0.449	0.000	B 49

[†]Computed on the 17 zero-shot outputs that state an ACR category.

Supplementary Table S4

Table S4: Validation of the assertion-level negation rule of Eq. 8. **(a)** Sensitivity / hallucination rate of the models of Table 6 on the leak-free subset ($n = 49$) as the negation window is varied; the 45-character column is the one used in the main text. Omission is 1−sensitivity. **(b)** Agreement between the 45-character rule and the ConText algorithm as implemented in medspaCy (default rules; target terms are the finding terms of the main-text rule), over all reference and generated reports of the five models. A mention is a single occurrence of a finding term, judged negated or not; a report-level assertion is whether a text asserts at least one finding, which is the quantity entering Eq. 8. All 25 disagreeing mentions on the full split were inspected: 24 are occurrences of the reference template "FINDINGS: No abnormal soft opacity or microcalcifications seen", which ConText does not negate after the section header (the same sentence without the header is negated), and the remaining one is a zero-shot mention preceded by "not" from the previous sentence, which does not change the report-level assertion. **(c)** lists every finding-category mismatch between the rule and the reference annotation (medspaCy ConText) on the leak-free subset, by text, finding category and direction; the 9 reference reports carrying the template are the 9 report-level disagreements of panel (b), and no mismatch runs in the opposite direction or involves asymmetry or architectural distortion. On the full split the same template adds 6 mentions from the 3 excluded images. **(d)** Finding-category level (mass, calcification, asymmetry, architectural distortion), which Eq. 8 does not measure: categories asserted by the generation but absent from the reference, the number of cases in which this happens, how many of those cases have a reference asserting no finding at all, and reference categories not asserted by the generation.

(a) Negation window

Method	**30 chars**	**45 chars**	**60 chars**	**90 chars**
MedGemma-4B (ZS)	0.375 / 0.061	0.300 / 0.000	0.300 / 0.000	0.300 / 0.000
DMID-only, r=16	0.575 / 0.000	0.575 / 0.000	0.575 / 0.000	0.575 / 0.000
Two-stage, r=16	0.525 / 0.000	0.525 / 0.000	0.525 / 0.000	0.525 / 0.000
DMID-only, r=64	0.675 / 0.000	0.675 / 0.000	0.675 / 0.000	0.675 / 0.000
Two-stage, r=64	0.650 / 0.000	0.650 / 0.000	0.650 / 0.000	0.650 / 0.000

(b) Agreement with medspaCy ConText

Set	**Unit**	n	**Agreement**	κ
Leak-free subset ($n = 49$)	finding mentions	705	97.3%	0.95
Leak-free subset ($n = 49$)	report-level assertions	294	96.9%	0.94
Full split ($n = 52$)	finding mentions	738	96.6%	0.93
Full split ($n = 52$)	report-level assertions	312	96.2%	0.92

Supplementary Table S4 (continued)

(c) Rule vs. ConText mismatches by finding category, leak-free subset

Direction / text	Images (Img)	Category	Terms	Mentions
Rule negates, ConText asserts				19
Reference	461, 463, 503–505, 507–510	mass	"opacity"	9
Reference	461, 463, 503–505, 507–510	calcification	"microcalcification(s)"	9
MedGemma-4B (ZS)	469	mass	"mass"†	1
Rule asserts, ConText negates				0
Total				19

†This single mismatch is an error of the rule—the negation cue of the preceding sentence falls within the 45-character window—and does not change the report-level assertion.

(d) Finding-category level, leak-free subset

Method	Extra categories	Cases	of which normal ref.	Missing categories
MedGemma-4B (ZS)	3/14	3	0	45/56
DMID-only, r=16	0/24	0	0	32/56
Two-stage, r=16	0/23	0	0	33/56
DMID-only, r=64	1/30	1	0	27/56
Two-stage, r=64	3/31	3	0	28/56

Supplementary Table S5

Table S5: Leave-one-out sensitivity of the headline metrics on the leak-free subset ($n = 49$). For each of the four fine-tuned models of Table 4, every metric is recomputed with each test case removed in turn; "Max. shift" is $\max_i |m_{-i} - m|$. Corpus-level metrics (BLEU, CIDEr) are recomputed at corpus level; BERTScore is recomputed per sample with the settings of the main text (roberta-large, no baseline rescaling). Paired ROUGE-L difference (two-stage − DMID-only): at $r = 16$ +0.0579, maximum leave-one-out shift 0.0039; at $r = 64$ +0.0010, maximum shift 0.0049.

	DMID-only, r=16		Two-stage, r=16		DMID-only, r=64		Two-stage, r=64	
Metric	Value	Max. shift	Value	Max. shift	Value	Max. shift	Value	Max. shift
BLEU-1	0.4323	0.0059	0.4879	0.0072	0.5324	0.0071	0.5203	0.0069
BLEU-4	0.1926	0.0042	0.2683	0.0056	0.3047	0.0059	0.3016	0.0059
ROUGE-1	0.6599	0.0052	0.7030	0.0065	0.7229	0.0068	0.7182	0.0067
ROUGE-L	0.5857	0.0049	0.6436	0.0059	0.6551	0.0061	0.6561	0.0061
METEOR	0.5776	0.0063	0.6377	0.0072	0.6729	0.0079	0.6664	0.0079
CIDEr	0.3618	0.0290	0.5559	0.0327	0.6119	0.0399	0.6495	0.0356
BERTScore	0.9065	0.0006	0.9137	0.0007	0.9179	0.0008	0.9175	0.0008
Sensitivity	0.5750	0.0147	0.5250	0.0135	0.6750	0.0173	0.6500	0.0167
BI-RADS exact	0.3265	0.0140	0.2857	0.0149	0.3265	0.0140	0.3265	0.0140